

CS & IT ENGINEERING

Programming in C

Control Flow Statements
Lec- 03



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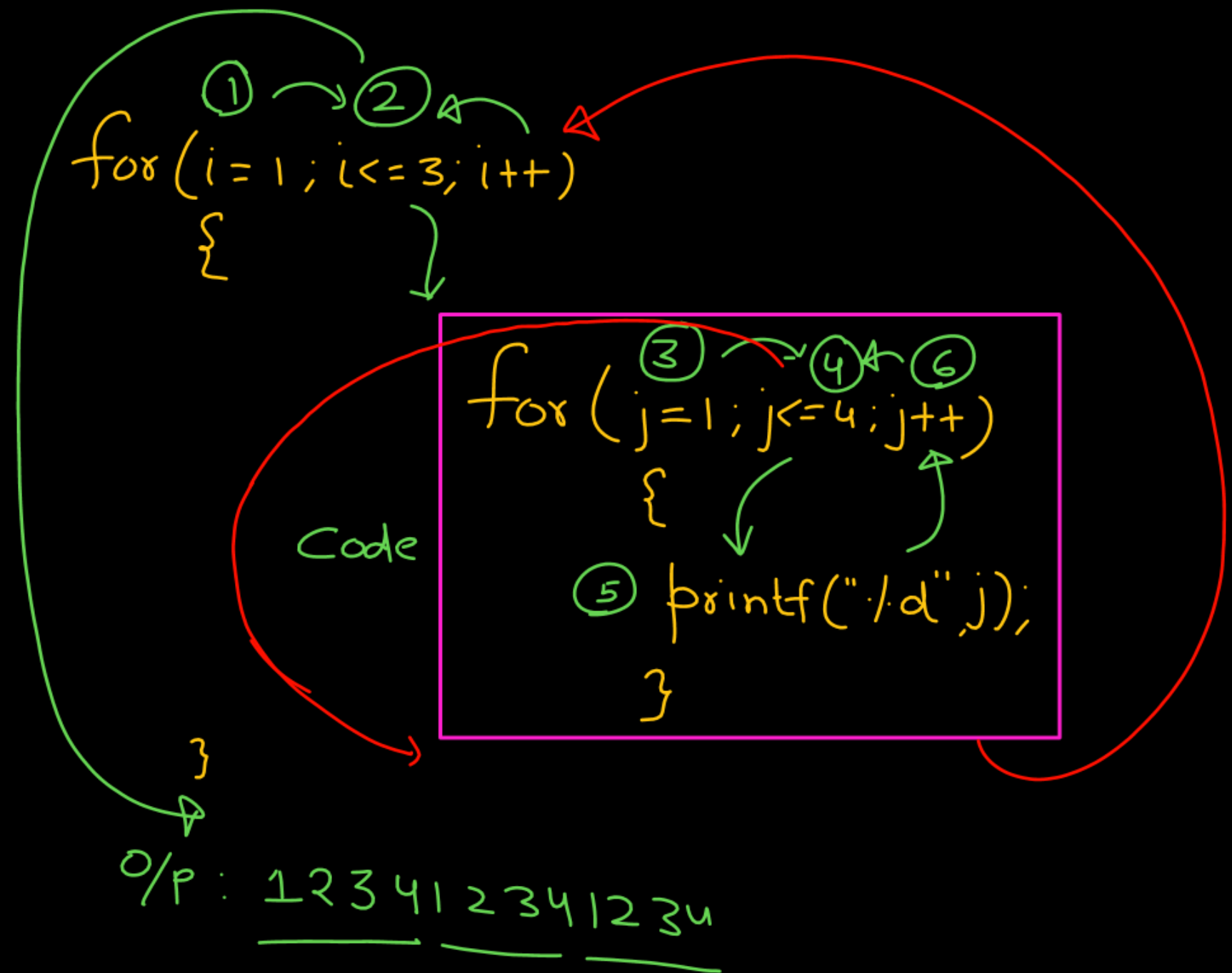


TOPICS TO BE
COVERED



Iterative Statements-II

i	Condition	
1	$1 \leq 3 \checkmark$	$j = 1, 2, 3, 4, \cancel{5}^x$
2	$2 \leq 3 \checkmark$	$j = 1, 2, 3, 4, \cancel{x}$
3	$3 \leq 3 \checkmark$	$j = 1, 2, 3, 4$
4	$4 \leq 3 \times$	



2.

i	
1	j=1,2,3,4,5, 6
2	j=1,2,3,4,5, 6
3	j=1,2,3,4,5, 6
4	

i=1 → Code ⇒ j=1,2,3,4,5
 i=2 → Code ⇒ j=1,2,3,4,5
 i=3 → Code ⇒ j=1,2,3,4,5

①
 for(i=1; i<=3; i++)
 {

Code

③ → ④ → ⑥
 for(j=1; j<=5; j++)
 {
 ⑤ printf("/d /d", i, j);
 }

1 1 1 2 1 3 1 4 1 5 2 1 2 2 2 3 2 4 2 5 3 1 3 2 3 3 3 4 3 5

3.

```
for (i=1; i<=3; i++)  
{
```

```
    for (j=1; j<=10; j++)  
    {
```

```
        printf("Pankaj");  
    }
```

```
}
```

i=1 → 10

i=2 → 10

i=3 → 10

10 + 10 + 10

= 3 × 10

= 30 times

4.

```
for (i = 1; i <= 3; i++)  
{  
    for (j = 1; j <= n; j++)  
    {  
        printf("Pankaj");  
    }  
}
```

i = 1 → n

i = 2 → n

i = 3 → n

n + n + n (3 times)

⇒ 3.n

5.

```
for (i=1; i<=10; i++)
```

$\Rightarrow 10$

```
{
```

```
for (j=1; j<=n; j++)
```

$\Rightarrow n$

```
{
```

```
printf("Pankaj");
```

```
}
```

```
}
```

$\Rightarrow 10 \times n$

= 10n times

iteration

S:

```
for (i=1; i<=n; i=i*2)
{
    for (j=1; j<=n; j++)
    {
        printf("Pankaj");
    }
}
```

$\rightarrow 1 + \lfloor \log_2 n \rfloor$

n

$i=1 \Rightarrow$

for each value of i

inner loop
executes
fixed no. of
times

$$\Rightarrow n(1 + \lfloor \log_2 n \rfloor)$$

6.

```
for (i=1; i<=n; i=i*2)
```

$\rightarrow 1 + \lfloor \log_2 n \rfloor$

```
{
```

```
  for (j=1; j<=n; j=j*2)
```

```
  {
```

```
    printf("Pankaj");
```

```
  }
```

```
}
```

$(1 + \lfloor \log_2 n \rfloor)$

$\Rightarrow (1 + \lfloor \log_2 n \rfloor)^2$

7

```
for (i = 1; i <= 3; i++)
```

```
{
```

```
    for (j = 1; j <= 3i; j++)  
    {  
        printf("Pankaj");  
    }
```

```
}
```

No. of times this
loop/code executes
depending upon ①

dependent nested loop

i	
1	j=1 → ①
2	j=1,2 → ②
3	j=1,2,3 → ③
1+2+3	
= 6	

$$\textcircled{1} \quad 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

$$\textcircled{2} \quad 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\textcircled{3} \quad 1^3 + 2^3 + 3^3 + \dots + n^3 = \left[\frac{n(n+1)}{2} \right]^2$$

A.P

$a, a+d, a+2d, \dots, a+(n-1)d$ n^{th} term

$$S_N = \frac{N}{2} [a_1 + l]$$

$$S_N = \frac{N}{2} [2a + (n-1)d]$$

G.P

$a, ar, ar^2, \dots, ar^{n-1}, \dots$

$$S_N = \frac{a(r^N - 1)}{r - 1}$$

$$S_N = \frac{a(1 - r^N)}{1 - r}$$

$$S_{\infty} = \frac{a}{1 - r}$$

Q

```
for (i=1; i<=n; i++)  
{  
    for (j=1; j<=3*i; j++)  
    {  
        printf("Panraj");  
    }  
}
```

i=1 j=1 to 3 3-1+1 ③	i=2 j=1 to 6 ⑥	i=3 j=1 to 9 ⑨	---
-------------------------------	----------------------	----------------------	-----

...	i=n j=1 to 3n ③n
-----	------------------------

$$3 + 6 + 9 + \dots + 3n$$

$$3(1 + 2 + 3 + \dots + n)$$

$$\frac{3n(n+1)}{2}$$

Q

```
for(i=1; i<=n; i++)  
{  
    for(j=i; j<=3*i; j++)  
    {  
        printf("Pankaj");  
    }  
}
```

A.P

first
↑

$i=1$	$i=2$	$i=3$	$i=4$
$j=1 \text{ to } 3$	$j=2 \text{ to } 6$	$j=3 \text{ to } 9$	$j=4 \text{ to } 12$
3	$6-2+1$ 5	$9-3+1$ 7	$12-4+1$ 9

$\dots$	$i=n$
	$j=n \text{ to } 3n$
	$3n-n+1$ $2n+1$

$3 + 5 + 7 + 9 + \dots + 2n+1$

$$= \frac{n}{2} [3 + 2n+1]$$

$$= \frac{n}{2} [2n+4] = \frac{n}{2} \times 2(n+2)$$

$$= n(n+2)$$

↘ Last term

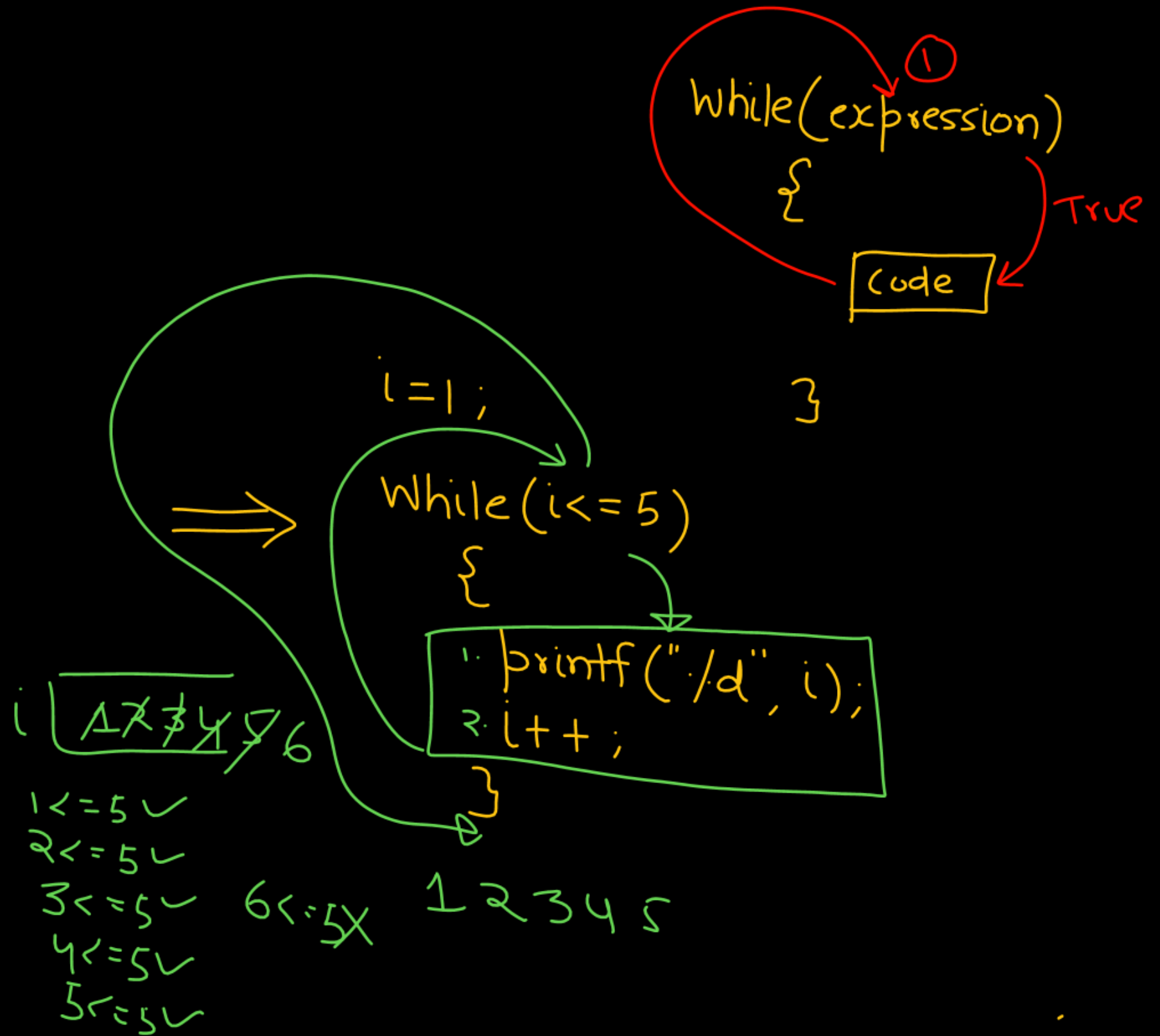
H.W

```
for (i = 1; i <= n; i = i * 3)
{
    for (j = i; j <= n; j++)
    {
        printf("Pankaj");
    }
}
```

Exact answer : ?

While loop

```
for(i=1; i<=5; i++)  
{  
    printf("/d", i);  
}
```



```
int i = 0;  
while(0i++)  
{
```

```
    printf("%d", i);  
}
```

No o/p

```
int i = -1;  
while(0++i){
```

```
    printf("%d", i);  
}
```

No o/p

for } pre-checking
while }

```
for( ; 0 ; )  
{  
  code  
}
```

```
while(0)  
{  
  code  
}
```

```
while(exp)  
{  
  while()  
  {  
    invalid  
  }  
}
```

do {

code

} while (exp) ;

do {

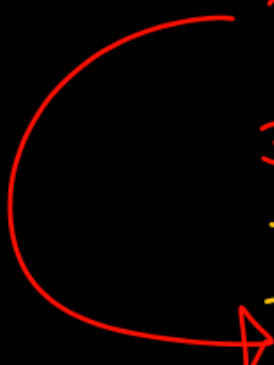
printf("Pankaj");

} while (0) ;

1 time

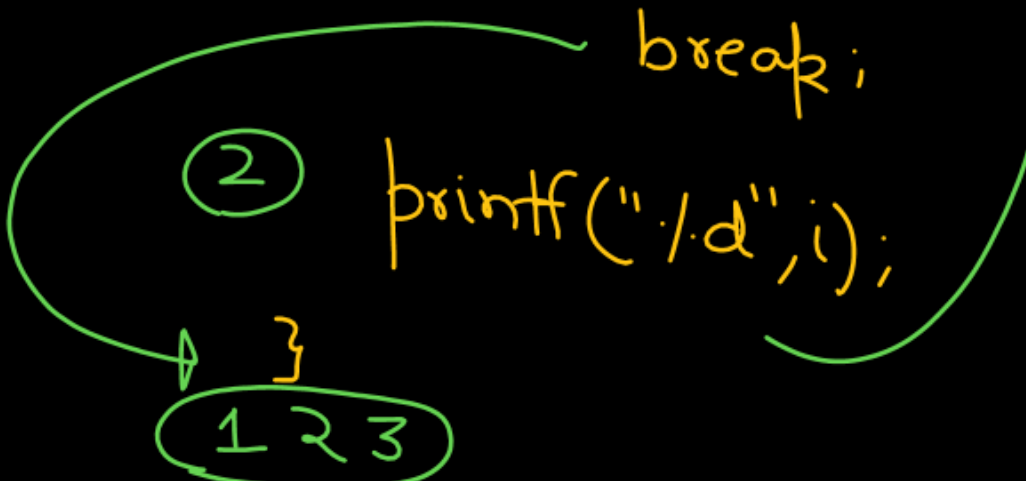
break :

```
for (i = 1; i <= 4; i++)  
{  
  1. printf("Pankaj");  
  2. break;  
  3. printf("/d", i);  
}
```



Pankaj

```
for (i = 1; i <= 10; i++)  
{  
  ① if (i / 4 == 0)  
    break;  
  ② printf("/d", i);  
}
```



i		
1	$1 <= 10 \checkmark$	$\Rightarrow 1 / 4 == 0 \times$
2	$2 <= 10 \checkmark$	$\Rightarrow 2 / 4 == 0 \times$
3	$3 <= 10 \checkmark$	$\Rightarrow 3 / 4 == 0 \times$
4	$4 <= 10 \checkmark$	$\Rightarrow 4 / 4 == 0 \checkmark$

```
for (i=1; i<=3; i++)  
{
```

i	
1	j=1,2,3
2	j=1,2
3	j=1
4	→ X

```
for (j=1; j<=4; j++)  
{
```

```
if ((i+j) % 4 == 0)
```

```
break;
```

```
printf("%d %d", i, j);
```

```
}
```

```
}
```

111221

Continue : skip the remaining portion of current iteration and continue with next iteration.

i		
1	1 <= 10 ✓	1 / 4 == 0 (F) → continue X
2	2 <= 10	2 / 4 == 0 (F) → continue X
3	3 <= 10	3 / 4 == 0 (F) → continue X
4	4 <= 10	4 / 4 == 0 (T) → continue
5	5 <= 10	5 / 4 == 0 (F) → cont X
6	6 <= 10	6 / 4 == 0 (F) → cont X
7	7 <= 10	7 / 4 == 0 (F) → cont X
8	8 <= 10	8 / 4 == 0 (T) → cont.
9	9 <= 10	9 / 4 == 0 (F) → cont X
10	10 <= 10	10 / 4 == 0 (F) → cont X

```
for(i=1; i<=10; i++)
{
```

```
1. if(i/4 == 0)
    continue;
```

```
2. printf("%d", i);
}
```

1 2 3 5 6 7 9 10

① `\n` → Move the cursor to the beginning of next line.

Hello Everyone_
How are you

`printf("Hello Everyone\n");`

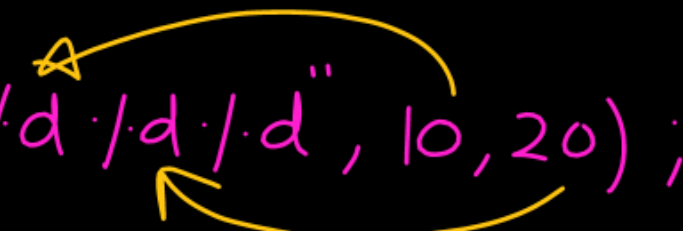
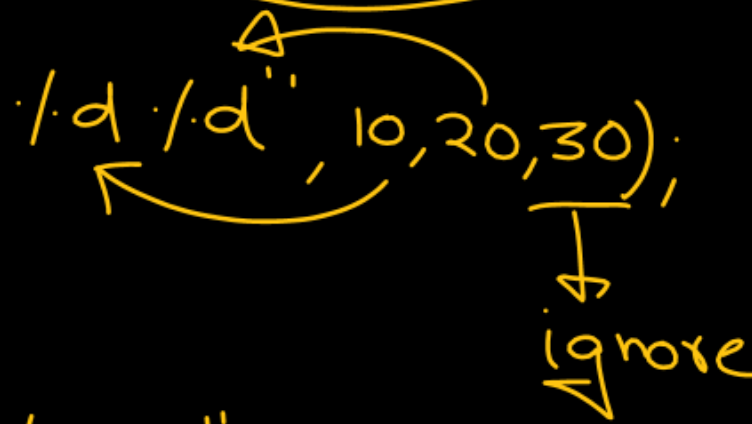
`printf("How are you");`

OR

`printf("Hello Everyone");`

`printf("\nHow are you");`

Hello Everyone_
How are you

- ① `printf("/d /d", 10, 20);`
- ② `printf("/d /d /d", 10, 20);` 10 20 Garbage

- ③ `printf("/d /d", 10, 20, 30);` 10 20

ignore
- ④ `printf("/d /d", 10, 20, 1);` ⇒ Error

